

13) (c)

$$2) \quad V_{aN} = V_{dc} \cdot d_1 + 0 \cdot (1-d_1) = V_{dc} d_1$$

$$(a) \quad V_{bN} = V_{dc} \cdot d_3 + 0 \cdot (1-d_3) = V_{dc} d_3$$

$$V_{cN} = V_{dc} \cdot d_5 + 0 \cdot (1-d_5) = V_{dc} d_5$$

$$\therefore V_{aN} = V_{dc} \left[\frac{1}{2} + \frac{m}{2} \sin(\omega t) + \frac{m}{12} \sin(3\omega t) \right]$$

$$V_{bN} = V_{dc} \left[\frac{1}{2} + \frac{m}{2} \sin\left(\omega t - \frac{2\pi}{3}\right) + \frac{m}{12} \sin(3\omega t) \right]$$

$$V_{cN} = V_{dc} \left[\frac{1}{2} + \frac{m}{2} \sin\left(\omega t + \frac{2\pi}{3}\right) + \frac{m}{12} \sin(3\omega t) \right]$$

$$V_{ab} = V_{aN} - V_{bN}$$

$$V_{ab} = V_{dc} \left[\frac{1}{2} + \frac{m}{2} \sin \omega t + \frac{m}{12} \sin 3\omega t - \frac{1}{2} - \frac{m}{2} \sin\left(\omega t - \frac{2\pi}{3}\right) - \frac{m}{12} \sin(3\omega t) \right]$$

$$V_{ab} = V_{dc} \left[\frac{m}{2} \sin(\omega t) - \frac{m}{2} \sin\left(\omega t - \frac{2\pi}{3}\right) \right]$$

$$V_{ab} = \frac{m}{2} V_{dc} \left[\sin \omega t - \sin\left(\omega t - \frac{2\pi}{3}\right) \right]$$

$$\boxed{V_{ab} = \sqrt{3} \frac{m}{2} V_{dc} \sin\left(\omega t + \frac{\pi}{6}\right)}$$

$$\text{ii) } \boxed{V_{bc} = V_{bN} - V_{cN} = \sqrt{3} \cdot \frac{m}{2} V_{dc} \sin\left(\omega t - \frac{\pi}{2}\right)}$$

$$\text{iii) } \boxed{V_{ca} = V_{cN} - V_{aN} = \sqrt{3} \cdot \frac{m}{2} V_{dc} \sin\left(\omega t + \frac{5\pi}{6}\right)}$$

(b)

let us take,

$$d_1 = \frac{1}{2} + \frac{m}{2} \sin(\omega t) + \frac{m}{12} \sin(3\omega t) \quad \text{--- (1)}$$

we need to find the value of m , such that d_1 saturates.

$$\therefore \frac{d}{d\omega t}(d_1) = \frac{m}{2} \cos(\omega t) + \frac{m}{4} \cos(3\omega t)$$

setting $\frac{d}{d\omega t}(d_1) = 0$ will give us extreme ^{points} ~~values~~ values.

$$\frac{m}{2} \cos(\omega t) + \frac{m}{4} \cos(3\omega t) = 0$$

$$\Rightarrow \boxed{2 \cos(\omega t) + \cos(3\omega t) = 0} \quad \text{--- (2)}$$

Plot the above function ⁽²⁾, we can see that

at $\omega t = 60^\circ, 90^\circ, 120^\circ, 240^\circ, 270^\circ, 300^\circ \Rightarrow \frac{d}{d\omega t}(d_1)$ is 0

ωt	$\frac{d^2(d_1)}{d(\omega t)^2}$	extrema type	d_1
60°	$-\frac{\sqrt{3}}{4} m$	maximum	1
90°	$+\frac{m}{4}$		
120°	$-\frac{\sqrt{3}}{4} m$	maximum	1
240°	$+\frac{\sqrt{3}}{4} m$	minimum	0
270°	$-\frac{m}{4}$		
300°	$+\frac{\sqrt{3}}{4} m$	minimum	0

$\therefore$ @ 60° in eq. (1) $d_1 = 1$ and $\omega t = 60^\circ \Rightarrow \boxed{m = \frac{2}{\sqrt{3}}}$

③ 6 switching are required to remove 5, 7, 11, 17 & 19 in

$$[V_{\text{switching node to AC bus-midpoint}}]$$

→ assume for a 3- ϕ , 2-level, 6 switches inverter,

in order to remove 5, 7, 11, 17 and 19 in line-to-line voltage, you will need only 6 switching $\therefore$ 3rd is zero-seq.

④ As derived in class,

$$f_o(\omega t) = \sum_{n=1}^{\infty} \left[\frac{4A}{n\pi} - \frac{8A}{n\pi} \cos(n\alpha_1) + \frac{8A}{n\pi} \cos(n\alpha_2) - \frac{8A}{n\pi} \cos(n\alpha_3) + \frac{8A}{n\pi} \cos(n\alpha_4) - \frac{8A}{n\pi} \cos(n\alpha_5) + \frac{8A}{n\pi} \cos(n\alpha_6) \dots \right] \sin(n\omega t)$$

$$f_o(\omega t) = \sum_{n=1}^{\infty} \frac{4A}{n\pi} \left[1 - 2\cos(n\alpha_1) + 2\cos(n\alpha_2) - 2\cos(n\alpha_3) + 2\cos(n\alpha_4) \right] \sin(n\omega t)$$

$$V_{ao} = \sum_{n=1}^{\infty} \frac{2V_{dc}}{n\pi} \left[1 - 2\cos(n\alpha_1) + 2\cos(n\alpha_2) - 2\cos(n\alpha_3) + 2\cos(n\alpha_4) \right] \sin(n\omega t)$$

$$V_{bo} = \sum_{n=1}^{\infty} \frac{2V_{dc}}{n\pi} \left[1 - 2\cos(n\alpha_1) + 2\cos(n\alpha_2) - 2\cos(n\alpha_3) + 2\cos(n\alpha_4) \right] \sin\left[n\left(\omega t - \frac{2\pi}{3}\right)\right]$$

$$V_{co} = \sum_{n=1}^{\infty} \frac{2V_{dc}}{n\pi} \left[1 - 2\cos(n\alpha_1) + 2\cos(n\alpha_2) - 2\cos(n\alpha_3) + 2\cos(n\alpha_4) \right] \sin\left[n\left(\omega t + \frac{2\pi}{3}\right)\right]$$

$$(V_{ao})_1 = \frac{2V_{dc}}{\pi} \times 0.9193 \sin(\omega t)$$

$$(V_{bo})_1 = \frac{2V_{dc}}{\pi} \times 0.9193 \sin\left(\omega t - \frac{2\pi}{3}\right)$$

$$(V_{co})_1 = \frac{2V_{dc}}{\pi} \times 0.9193 \sin\left(\omega t + \frac{2\pi}{3}\right)$$

$$(V_{ao})_3 = \frac{2V_{dc}}{3\pi} \times 0.4245 \sin(3\omega t)$$

$$(V_{bo})_3 = \frac{2V_{dc}}{3\pi} \times 0.4245 \sin(3\omega t)$$

$$(V_{co})_3 = \frac{2V_{dc}}{3\pi} \times 0.4245 \sin(3\omega t)$$

$$\begin{array}{c|c|c|c} (V_{ao})_5 \approx 0 & (V_{ao})_7 \approx 0 & (V_{ao})_{11} \approx 0 & (V_{ao})_{13} \approx 0 \\ (V_{bo})_5 \approx 0 & (V_{bo})_7 \approx 0 & (V_{bo})_{11} \approx 0 & (V_{bo})_{13} \approx 0 \\ (V_{co})_5 \approx 0 & (V_{co})_7 \approx 0 & (V_{co})_{11} \approx 0 & (V_{co})_{13} \approx 0 \end{array}$$

$$(V_{ab})_1 = \sqrt{3} \times \frac{2V_{dc}}{\pi} \times 0.9193 \sin\left(\omega t + \frac{\pi}{6}\right)$$

$$(V_{ab})_1 = (1.01367 V_{dc}) \sin\left(\omega t + \frac{\pi}{6}\right)$$

$$(V_{bc})_1 = (1.01367 V_{dc}) \sin\left(\omega t - \frac{\pi}{2}\right)$$

$$(V_{ca})_1 = (1.01367 V_{dc}) \sin\left(\omega t + \frac{5\pi}{6}\right)$$

$$(V_{ab})_{3,5,7,11,13} = 0$$

$$(V_{bc})_{3,5,7,11,13} = 0$$

$$(V_{ca})_{3,5,7,11,13} = 0$$

5) b, c, d

6)

a)

$$|V_a| = \frac{2}{\sqrt{3}} |V^*| \sin(60^\circ - \theta)$$

$$|V_b| = \frac{2}{\sqrt{3}} |V^*| \sin \theta$$

$$\theta = r - \left[(s-1) \frac{\pi}{3} \right]$$

$$\theta = 15 - \left[(1-1) \frac{\pi}{3} \right] = 15^\circ$$

$$\theta = r - \left[(s-1) \frac{\pi}{3} \right]$$

we need, $V^* = 0.5 V_{dc} \angle 15^\circ$

here, $r = 15^\circ$, $s = \text{sector number}$

so, in this case, $r = 15^\circ$, $s = 1$

$$|V_a| = \frac{2}{\sqrt{3}} \times 0.5 V_{dc} \times \sin(60^\circ - 15^\circ) = 0.4083 V_{dc}$$

$$|V_b| = \frac{2}{\sqrt{3}} \times 0.5 V_{dc} \times \sin(15^\circ) = 0.1495 V_{dc}$$

as w.k.T we are in sector-1 bounding states are

$$V_1 \Rightarrow (1, 0, 0) \rightarrow \sqrt{\frac{2}{3}} V_{dc} \angle 0^\circ$$

$$V_2 \Rightarrow (1, 1, 0) \rightarrow \sqrt{\frac{2}{3}} V_{dc} \angle 60^\circ$$

$$t_a = \frac{|V_a|}{|V_1|} \cdot T_s = \frac{0.4083 V_{dc}}{\sqrt{\frac{2}{3}} V_{dc}} \times 50 \mu = 25.0032 \mu s$$

$$t_b = \frac{|V_b|}{|V_2|} \cdot T_s = \frac{0.1495 V_{dc}}{\sqrt{\frac{2}{3}} V_{dc}} \times 50 \mu = 9.155 \mu s$$

$$t_0 = T_s - (t_a + t_b) = 15.8418 \mu s$$

$$(t_{on})_{s1} = t_a + t_b + \frac{t_o}{2} + \frac{t_o}{2} + t_a + t_b = 84.1582 \mu s$$

$$(t_{on})_{s3} = t_b + \frac{t_o}{2} + \frac{t_o}{2} + t_b = 34.1518 \mu s$$

$$(t_{on})_{s5} = \frac{t_o}{2} + \frac{t_o}{2} = 15.8418 \mu s.$$

(b) here we need $v^* = 0.5 V_{dc} \angle 150^\circ$

where, $r = 150^\circ$, the space vector is in sector 3

$$\theta = r - \left[(s-1) \frac{\pi}{3} \right] = 150 - ((3-1) \times 60) = \underline{30^\circ}$$

$$|V_a| = \frac{2}{\sqrt{3}} \times 0.5 \times V_{dc} \times \sin(60-30) = 0.2887 V_{dc}$$

$$|V_b| = \frac{2}{\sqrt{3}} \times 0.5 \times V_{dc} \times \sin(30^\circ) = 0.2887 V_{dc}$$

as, w.r.t we are in sector 3, the bounding states are,

$$V_1 \rightarrow (0, 1, 0) \rightarrow \sqrt{\frac{2}{3}} V_{dc} \angle 120^\circ$$

$$V_2 \rightarrow (0, 1, 1) \rightarrow \sqrt{\frac{2}{3}} V_{dc} \angle 180^\circ$$

$$t_a = \frac{|V_a|}{|V_1|} \cdot T_s = \frac{0.2887 V_{dc}}{\sqrt{\frac{2}{3}} V_{dc}} \times 50 \mu s = 17.6792 \mu s$$

$$t_b = \frac{|V_b|}{|V_2|} \cdot T_s = \frac{0.2887 V_{dc}}{\sqrt{\frac{2}{3}} V_{dc}} \times 50 \mu s = 17.6792 \mu s$$

$$t_o = T_s - (t_a + t_b) = 14.6416 \mu s.$$

$$(t_{on})_{s1} = \frac{t_o}{2} + \frac{t_o}{2} = 14.6416 \mu s$$

$$(t_{on})_{s3} = t_a + t_b + \frac{t_o}{2} + \frac{t_o}{2} + t_b + t_a = 85.3584 \mu s$$

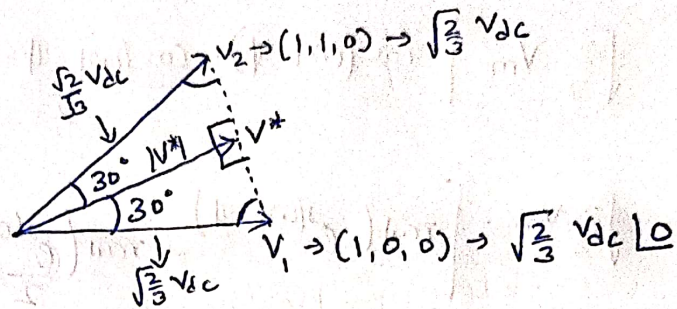
$$(t_{on})_{s5} = t_b + \frac{t_o}{2} + \frac{t_o}{2} + t_b = 50 \mu s.$$

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as the space vector is at an angle 30° , w.r.t it is in

Sector 1

space vector tip just touches the hexagonal boundary.



$$\therefore \cos 30^\circ = \frac{|V_2|}{|V^*|} \Rightarrow \frac{2}{\sqrt{3}} = \frac{\frac{\sqrt{2}}{\sqrt{3}} V_{dc}}{|V^*|} \Rightarrow \boxed{|V^*| = \frac{V_{dc}}{\sqrt{2}}}$$

$$\therefore |V_a| = \frac{2}{\sqrt{3}} \times \frac{V_{dc}}{\sqrt{2}} \times \sin(60-30) = \frac{V_{dc}}{\sqrt{6}}$$

$$|V_b| = \frac{2}{\sqrt{3}} \times \frac{V_{dc}}{\sqrt{2}} \times \sin 30 = \frac{V_{dc}}{\sqrt{6}}$$

$$\therefore t_a = \frac{|V_a|}{|V_1|} T_s = \frac{\left(\frac{V_{dc}}{\sqrt{6}}\right)}{\frac{\sqrt{2}}{\sqrt{3}} V_{dc}} \cdot 50\mu = 25\mu$$

$$t_b = \frac{|V_b|}{|V_1|} T_s = \frac{\left(\frac{V_{dc}}{\sqrt{6}}\right)}{\frac{\sqrt{2}}{\sqrt{3}} V_{dc}} \cdot 50\mu = 25\mu$$

$$\boxed{t_0 = 0\mu s.}$$

8

$$V_a = 325 \cos(\omega t - 22.5^\circ) = V_m \cos(\omega t - \phi)$$

$$V_b = 325 \cos(\omega t - 142.5^\circ) = V_m \cos(\omega t - \phi - 120^\circ)$$

$$V_c = 325 \cos(\omega t - 262.5^\circ) = V_m \cos(\omega t - \phi - 240^\circ)$$

$$\begin{bmatrix} V_\alpha \\ V_\beta \\ V_0 \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

$$V_d = \sqrt{\frac{2}{3}} \left[V_a - \frac{V_b}{2} - \frac{V_c}{2} \right]$$

$$V_d = \sqrt{\frac{2}{3}} V_m \left[\cos(\omega t - \phi) - \frac{\cos(\omega t - \phi - 120^\circ)}{2} - \frac{\cos(\omega t - \phi - 240^\circ)}{2} \right]$$

$$V_d = \sqrt{\frac{2}{3}} V_m \left[\text{real} \left(e^{j(\omega t - \phi)} \right) - \text{real} \left(\frac{e^{j(\omega t - \phi - 120^\circ)}}{2} \right) - \text{real} \left(\frac{e^{j(\omega t - \phi - 240^\circ)}}{2} \right) \right]$$

$$V_d = \sqrt{\frac{2}{3}} V_m \times \text{real} \left[e^{j\omega t} \cdot \left[e^{-j\phi} - \frac{e^{-j(\phi + 120^\circ)}}{2} - \frac{e^{-j(\phi + 240^\circ)}}{2} \right] \right]$$

$$V_d = \sqrt{\frac{2}{3}} V_m \cdot \text{real} \left[e^{j\omega t} \cdot e^{-j\phi} \underbrace{\left[1 - \frac{e^{-j120^\circ}}{2} - \frac{e^{-j240^\circ}}{2} \right]}_{\downarrow \frac{3}{2}} \right]$$

$$V_d = \sqrt{\frac{2}{3}} V_m \cdot \text{real} \left[\frac{3}{2} \cdot e^{j(\omega t - \phi)} \right] = \sqrt{\frac{3}{2}} V_m \text{real} \left[e^{j(\omega t - \phi)} \right]$$

$$\boxed{V_d = \sqrt{\frac{3}{2}} V_m \cos(\omega t - \phi)}$$

$$V_\beta = \sqrt{\frac{2}{3}} \left[\frac{\sqrt{3}}{2} V_b - \frac{\sqrt{3}}{2} V_c \right] = \frac{V_m}{\sqrt{2}} \left[\cos(\omega t - \phi - 120^\circ) - \cos(\omega t - \phi - 240^\circ) \right]$$

$$V_\beta = \frac{V_m}{\sqrt{2}} \left[\text{real} \left(e^{j(\omega t - \phi - 120^\circ)} \right) - \text{real} \left(e^{j(\omega t - \phi - 240^\circ)} \right) \right]$$

$$V_\beta = \frac{V_m}{\sqrt{2}} \text{real} \left[e^{j(\omega t - \phi)} \cdot \underbrace{\left(e^{-j120^\circ} - e^{-j240^\circ} \right)}_{\downarrow \sqrt{3} \cdot e^{-j90^\circ}} \right]$$

$$V_\beta = \frac{V_m}{\sqrt{2}} \text{real} \left[e^{j(\omega t - \phi)} \cdot \sqrt{3} \cdot e^{-j90^\circ} \right]$$

$$V_{\beta} = \frac{V_m}{\sqrt{2}} \operatorname{real} \left[e^{j(\omega t - \phi)} \cdot \sqrt{3} \cdot e^{-j90^\circ} \right]$$

$$V_{\beta} = \sqrt{\frac{3}{2}} V_m \operatorname{real} \left[e^{j(\omega t - \phi - 90^\circ)} \right]$$

$$V_{\beta} = \sqrt{\frac{3}{2}} V_m \cos(\omega t - \phi - 90^\circ)$$

$$\boxed{V_{\beta} = \sqrt{\frac{3}{2}} V_m \sin(\omega t - \phi)}$$

$$\therefore \text{space/time phasor } V^* = V_{\alpha} + j V_{\beta}$$

$$V^* = \left[\sqrt{\frac{3}{2}} V_m \cos(\omega t - \phi) + \left[j \cdot \sqrt{\frac{3}{2}} V_m \sin(\omega t - \phi) \right] \right]$$

$$V^* = \sqrt{\frac{3}{2}} V_m \left[\cos(\omega t - \phi) + j \sin(\omega t - \phi) \right]$$

$$\boxed{V^* = \sqrt{\frac{3}{2}} V_m e^{j(\omega t - \phi)}}$$

now, w.k.T $\phi = 22.5^\circ$, $V_m = 325 \text{ V}$, and it is asked to

find out space vector at $t = 12 \text{ ms}$, given, $\omega = 100 \pi \text{ (rad/s)}$

$$\therefore V^* = \sqrt{\frac{3}{2}} \times 325 e^{j \left[(100 \pi \times 10^{-3} \times 12) \times \frac{180}{\pi} \right] - 22.5^\circ}$$

$$V^* = \sqrt{\frac{3}{2}} \times 325 e^{j(216^\circ - 22.5^\circ)}$$

$$\boxed{V^* = 348.0421 e^{j193.5^\circ}}$$

as the angle is 193.5° , the space vector is in sector "4"
and the bounding states are $V_1 \rightarrow (0, 1, 1)$ & $V_2 \rightarrow (0, 0, 1)$.

(4) we, have already derived 'a' component, "B" component in prob (3).

(a)
$$\begin{aligned} i_a &= \sqrt{\frac{2}{3}} I_m \cos(\omega t - \phi) \\ i_b &= \sqrt{\frac{2}{3}} I_m \sin(\omega t - \phi) \end{aligned}$$

$$i_o = \sqrt{\frac{2}{3}} \left[\frac{I_m}{\sqrt{2}} \cos(\omega t - \phi) + \frac{I_m}{\sqrt{2}} \cos(\omega t - \phi - 120^\circ) + \frac{I_m}{\sqrt{2}} \cos(\omega t - \phi - 240^\circ) \right]$$

$$i_o = \sqrt{\frac{2}{3}} \frac{I_m}{\sqrt{2}} \underbrace{[\cos(\omega t - \phi) + \cos(\omega t - \phi - 120^\circ) + \cos(\omega t - \phi - 240^\circ)]}_{\downarrow 0}$$

$$\boxed{i_o = 0}$$

$$\therefore \left. \begin{aligned} i_a &= \sqrt{\frac{2}{3}} I_m \cos(\omega t - \phi) \\ i_b &= \sqrt{\frac{2}{3}} I_m \sin(\omega t - \phi) \\ i_o &= 0 \end{aligned} \right\} \begin{array}{l} \text{These equations hold} \\ \text{true ONLY when} \\ i_a, i_b \text{ \& } i_c \text{ are} \\ \text{balanced} \end{array}$$

$$\therefore i_a = \sqrt{\frac{2}{3}} \times 5 \times \cos(\omega t - \phi) = 6.1237 \cos(\omega t - \phi)$$

$$i_b = \sqrt{\frac{2}{3}} \times 5 \times \sin(\omega t - \phi) = 6.1237 \sin(\omega t - \phi)$$

$$i_o = 0$$

(b)

$$\begin{bmatrix} i_a \\ i_b \\ i_o \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}$$

$$i_a = \sqrt{\frac{2}{3}} \left(i_a - \frac{i_b}{2} - \frac{i_c}{2} \right)$$

$$i_a = \sqrt{\frac{2}{3}} \left[5 \cos(\omega t - \phi) - \frac{4.9}{2} \cos(\omega t - \phi + 120^\circ) - \frac{5}{2} \cos(\omega t - \phi + 240^\circ) \right]$$

$$i_a = \sqrt{\frac{2}{3}} \left[5 \operatorname{real} \left(e^{j(\omega t - \phi)} \right) - 2.45 \operatorname{real} \left(e^{j(\omega t - \phi + 120^\circ)} \right) - 2.5 \operatorname{real} \left(e^{j(\omega t - \phi + 240^\circ)} \right) \right]$$

$$i_a = \sqrt{\frac{2}{3}} \times \operatorname{real} \left[e^{j(\omega t - \phi)} \left(5 - 2.45 e^{+j120} - 2.5 e^{+j240} \right) \right]$$

$$i_a = \sqrt{\frac{2}{3}} \operatorname{real} \left[1 \angle (\omega t - \phi) \left(5 - 2.45 \angle 120^\circ - 2.5 \angle 240^\circ \right) \right]$$

$$i_a = \sqrt{\frac{2}{3}} \operatorname{real} \left[1 \angle (\omega t - \phi) \times 7.4751 \angle 0.3319 \right]$$

$$i_a = \sqrt{\frac{2}{3}} \operatorname{real} \left[7.4751 \angle (\omega t - \phi + 0.3319) \right]$$

$$i_a = \sqrt{\frac{2}{3}} \operatorname{real} \left[7.4751 e^{j(\omega t - \phi + 0.3319)} \right]$$

$$i_a = \sqrt{\frac{2}{3}} \cdot 7.475 \cos(\omega t - \phi + 0.3319)$$

$$\boxed{i_a = 6.1033 \cos(\omega t - \phi + 0.3319)}$$

$$i_B = \sqrt{\frac{2}{3}} \left[\frac{\sqrt{3}}{2} i_b - \frac{\sqrt{3}}{2} i_c \right]$$

$$i_B = \frac{1}{\sqrt{2}} [i_b - i_c]$$

$$i_B = \frac{1}{\sqrt{2}} \left[4.9 \cos(\omega t - \phi + 120^\circ) - 5 \cos(\omega t - \phi + 240^\circ) \right]$$

$$i_B = \frac{1}{\sqrt{2}} \left[4.9 \operatorname{real} \left(e^{j(\omega t - \phi + 120^\circ)} \right) - 5 \operatorname{real} \left(e^{j(\omega t - \phi + 240^\circ)} \right) \right]$$

$$i_B = \frac{1}{\sqrt{2}} \left[\operatorname{real} \left(e^{j(\omega t - \phi)} \cdot (4.9 e^{j120^\circ} - 5 e^{j240^\circ}) \right) \right]$$

$$i_B = \frac{1}{\sqrt{2}} \times \operatorname{real} \left[1 \angle \omega t - \phi \left(4.9 \angle 120^\circ - 5 \angle 240^\circ \right) \right]$$

$$i_B = \frac{1}{\sqrt{2}} \operatorname{real} \left[(1 \angle \omega t - \phi) \times (8.5738 \angle 89.6659^\circ) \right]$$

$$i_B = \frac{1}{\sqrt{2}} \operatorname{real} \left[8.5738 \angle \omega t - \phi + 89.6659^\circ \right]$$

$$i_B = \frac{1}{\sqrt{2}} \operatorname{real} \left[8.5738 e^{j(\omega t - \phi + 89.6659^\circ)} \right]$$

$$i_B = \frac{1}{\sqrt{2}} \times 8.5738 \cos(\omega t - \phi + 89.6659^\circ)$$

$$i_B = 6.0626 \cos(90^\circ + \omega t - \phi - 0.3341^\circ)$$

$$i_B = -6.0626 \sin(\omega t - \phi - 0.3341^\circ)$$

$$i_0 = \sqrt{\frac{2}{3}} \times \left(\frac{i_a}{\sqrt{2}} + \frac{i_b}{\sqrt{2}} + \frac{i_c}{\sqrt{2}} \right)$$

$$i_0 = \frac{1}{\sqrt{3}} \left[5 \cos(\omega t - \phi) + 4.9 \cos(\omega t - \phi + 120^\circ) + 5 \cos(\omega t - \phi + 240^\circ) \right]$$

$$i_0 = \frac{1}{\sqrt{3}} \left[5 \operatorname{real} \left(e^{j(\omega t - \phi)} \right) + 4.9 \operatorname{real} \left(e^{j(\omega t - \phi + 120^\circ)} \right) + 5 \operatorname{real} \left(e^{j(\omega t - \phi + 240^\circ)} \right) \right]$$

$$i_0 = \frac{1}{\sqrt{3}} \times \text{real} \left[e^{j(\omega t - \phi)} \cdot [5 + 4.9 e^{j120^\circ} + 5 e^{j240^\circ}] \right]$$

$$i_0 = \frac{1}{\sqrt{3}} \text{real} \left[1 \angle (\omega t - \phi) [5 + 4.9 \angle 120^\circ + 5 \angle 240^\circ] \right]$$

$$i_0 = \frac{1}{\sqrt{3}} \text{real} \left[1 \angle (\omega t - \phi) \times 0.1 \angle -60^\circ \right]$$

$$i_0 = \frac{1}{\sqrt{3}} \text{real} \left[0.1 \angle (\omega t - \phi - 60^\circ) \right]$$

$$i_0 = \frac{1}{\sqrt{3}} \text{real} \left[0.1 e^{j(\omega t - \phi - 60^\circ)} \right]$$

$$i_0 = \frac{1}{\sqrt{3}} \times 0.1 \cos (\omega t - \phi - 60^\circ)$$

$$i_0 = 0.05774 \cos (\omega t - \phi - 60^\circ)$$

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